

SAMAIL, Oman

WMO: 412570

Lat: 23.254N

Lon: 57.928E

Elev: 417

StdP: 96.41

Time Zone: 4.00 (E04)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.9	14.2	-4.8	2.6	32.1	-2.4	3.2	30.8	8.5	22.7	7.2	21.6	0.8	210	0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	10.3	44.6	21.3	43.6	21.6	42.7	21.8	27.5	36.5	26.8	36.0	26.1	35.7	4.6	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	21.5	33.1	24.3	20.3	33.1	23.3	19.1	33.1	90.5	36.3	87.1	36.3	83.9	35.6	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.1	6.7	6.0	9.8	46.6	1.3	0.6	8.8	47.0	8.1	47.4	7.4	47.8	6.4	48.2		
			5.6	28.5	1.6	1.0	4.5	29.1	3.5	29.7	2.6	30.2	1.4	30.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.7	20.1	22.4	26.1	30.6	35.2	37.3	37.3	36.2	34.1	30.5	25.2	21.1
	DBStd	6.59	2.22	2.80	2.83	2.73	2.24	1.94	2.00	1.97	1.62	1.88	2.16	2.06
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	14	8	2	0	0	0	0	0	0	0	0	0	4
	CDD10.0	7196	313	347	500	617	781	819	845	812	724	634	457	346
	CDD18.3	4168	63	116	242	367	523	569	587	553	474	376	207	91
	CDH23.3	61691	311	844	2517	5119	8678	9886	10091	9243	7507	5167	1838	491
	CDH26.7	41359	47	262	1157	3017	6221	7489	7612	6763	5114	2942	658	77
(14) Wind	WSAvg	2.5	2.0	2.4	2.4	2.7	3.0	3.1	3.0	2.7	2.6	2.2	1.9	1.8
(15) Precipitation	PrecAvg	170	11	18	27	20	10	9	20	15	11	9	9	10
	PrecMax	592	72	95	307	74	58	64	106	65	67	52	67	57
	PrecMin	57	0	0	0	0	0	0	1	0	0	0	0	1
	PrecStd	97	19	22	58	22	12	17	26	15	14	13	17	13
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.1	33.7	36.7	40.2	44.9	46.0	45.6	44.7	42.2	39.5	34.2	30.1
		MCWB	15.7	17.1	18.0	19.0	20.1	20.1	21.7	22.0	21.1	19.0	18.2	17.0
	2%	DB	27.9	31.4	35.3	39.0	43.1	44.7	44.3	43.4	41.1	38.3	32.6	28.6
		MCWB	16.4	17.3	17.7	18.8	20.6	20.6	22.1	22.3	21.4	19.2	18.3	17.3
	5%	DB	26.4	29.7	33.9	37.8	41.7	43.7	43.3	42.2	40.2	37.0	31.5	27.3
		MCWB	16.5	17.1	17.5	18.9	20.4	21.0	22.5	22.4	21.3	19.2	18.4	17.3
	10%	DB	25.0	28.1	32.3	36.5	40.6	42.6	42.3	41.2	39.1	35.8	30.3	26.1
		MCWB	16.2	16.9	17.4	18.6	20.0	21.2	22.8	22.8	21.3	19.0	18.3	17.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.5	21.1	22.4	24.0	25.8	27.6	28.6	28.1	26.8	24.9	22.6	20.7
		MCDB	25.0	27.2	28.3	31.0	33.2	37.1	37.7	36.6	34.3	31.2	28.7	24.8
	2%	WB	18.7	19.7	21.2	22.8	24.7	26.5	27.6	27.4	25.8	23.5	21.6	19.9
		MCDB	23.7	25.9	27.5	29.9	33.2	36.0	37.0	35.9	34.1	31.2	27.4	24.3
	5%	WB	18.0	19.0	20.4	21.9	23.9	25.7	26.8	26.7	24.9	22.5	20.8	19.2
		MCDB	23.0	25.1	27.6	29.7	33.7	35.7	36.9	35.5	33.9	30.8	26.9	23.8
	10%	WB	17.3	18.3	19.6	21.1	23.1	24.9	26.0	26.1	24.1	21.7	20.2	18.6
		MCDB	22.6	24.9	27.6	30.8	34.2	35.7	36.5	35.4	34.3	30.8	26.6	23.4
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	10.5	10.9	10.4	10.4	10.3	9.7	9.9	10.3	10.6	9.7	9.6
		MCDBR	11.2	12.1	12.4	11.3	11.2	11.2	10.7	10.5	10.7	11.5	10.5	11.1
		MCWBR	4.1	4.6	4.7	4.1	4.6	4.3	4.0	3.9	4.3	4.6	4.6	4.3
	5% WB	MCDBR	9.7	9.8	9.7	9.6	10.3	9.6	8.8	9.1	9.7	9.7	8.9	8.9
		MCWBR	4.2	4.2	4.7	4.5	5.3	5.1	4.2	3.8	4.1	4.6	4.4	3.7
(40) Clear-Sky Solar Irradiance	taub		0.396	0.439	0.502	0.560	0.590	0.660	0.736	0.678	0.569	0.462	0.420	0.396
	taud		2.261	2.104	1.916	1.810	1.748	1.637	1.543	1.632	1.813	2.049	2.197	2.251
	Ebn at Noon		859	843	806	767	739	684	634	671	739	808	825	841
	Edh at Noon		123	153	194	219	232	257	283	259	212	160	130	120
(44) All-Sky Solar Radiation	RadAvg		4.60	5.40	6.26	6.92	7.53	7.36	6.99	6.71	6.35	5.78	4.81	4.35
	RadStd		0.21	0.24	0.33	0.37	0.17	0.26	0.28	0.17	0.12	0.16	0.14	0.21

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	+0.42	+0.68	N/A	N/A	+0.28	+0.30	N/A	N/A	+96	+98

Nomenclature: See separate page